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The Prevention and Treatment of Exercise-Induced Muscle Damage

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Contents

Abstract

Exercise-induced muscle damage (EIMD) can be caused by novel or unaccustomed exercise and results in a temporary decrease in muscle force production, a rise in passive tension, increased muscle soreness and swelling, and an increase in intramuscular proteins in blood. Consequently, EIMD can have a profound effect on the ability to perform subsequent bouts of exercise and therefore adhere to an exercise training programme. A variety of interventions have been used prophylactically and/or therapeutically in an attempt to reduce the negative effects associated with EIMD. This article focuses on some of the most commonly used strategies, including nutritional and pharmacological strategies, electrical and manual therapies and exercise. Long-term supplementation with antioxidants or β-hydroxy-β-methylbutyrate appears to provide a prophylactic effect in reducing EIMD, as does the ingestion of protein before and following exercise. Although the administration of high-dose NSAIDs may reduce EIMD and muscle soreness, it also attenuates the adaptive processes and should therefore not be prescribed for long-term treatment of EIMD. Whilst there is some evidence that stretching and massage may reduce muscle soreness, there is little evidence indicating any performance benefits. Electrical therapies and cryotherapy offer limited effect in the treatment of EIMD; however, inconsistencies in the dose and frequency of these and other interventions may account for the lack of consensus regarding their efficacy. Both as a cause and a consequence of this, there are very few evidence-based guidelines for the application of many of these interventions. Conversely, there is unequivocal evidence that prior bouts of eccentric exercise provide a protective effect against subsequent bouts of potentially damaging exercise. Further research is warranted to elucidate the most appropriate dose and frequency of interventions to attenuate EIMD and if these interventions attenuate the adaptation process. This will both clarify the efficacy of such strategies and provide guidelines for evidence-based practice.

Eccentric contractions occur when an external force is greater than that being exerted by the muscle and hence lead to the muscle lengthening, whilst tension is generated.[1] In addition, the torque generated by maximal eccentric contractions is sizeable and may greatly exceed maximal voluntary isometric contractions.[2] Although the torque produced during eccentric contractions far exceeds that produced in other contraction types,[3] the metabolic cost and neural activation is less per unit of torque.[4-7]

Exercise-induced muscle damage (EIMD) is a phenomenon that occurs as a result of novel or unaccustomed exercise; the severity of the damage and the extent of discomfort are exacerbated over time and can last for several days when the exercise bout encompasses an eccentric component.[8,9] This damage manifests itself as a temporary decrease in muscle function (force production and rise in passive tension), increased muscle soreness, an increased swelling of the involved muscle group and an increase in intramuscular proteins in blood. Although these variables are commonly used to assess EIMD, the methods employed can vary considerably between investigations and have been reviewed elsewhere.[10] Most damage occurs when the exercise bout is both novel and eccentric in nature. Although eccentric contractions are potentially damaging, there are a number of benefits in conducting this muscle action; they produce greater gains in hypertrophy than concentric contractions[3,6,11-13] and increase eccentric contraction-specific strength.[6,14] In addition, eccentric work occurs at a lower metabolic cost than does concentric work and therefore a lower cardiovascular stress is involved, which makes this mode of exercise attractive to populations such as the elderly, who retain greater levels of eccentric strength with age, and those with cardiovascular pathologies.[15,16]

The culmination of benefits from eccentric muscle actions provide compelling rationale for their inclusion in exercise and training regimens; however, the damaging effects of eccentric muscle actions, thought necessary for adaptive remodelling,[17,18] can affect subsequent exercise sessions due to residual muscle pain, restriction of movement and reduced capacity to exercise at an intensity that may be beneficial for the exerciser. Therefore, any intervention that may alleviate the negative symptoms of temporary EIMD may be beneficial to a range of exercising populations, from elite athletes to the elderly and chronically diseased. A number of investigations have examined various interventions in an attempt to reduce these negative effects. These investigations vary considerably in exercise modality, muscle group used, duration and frequency of treatments and population size, which can make conclusions sometimes difficult to make. This review is not exhaustive; however, its purpose is to provide the reader with a current overview of the literature that examines interventions administered as a prophylactic or as a therapy to alleviate the negative effects of EIMD.

1. Proposed Mechanisms of Muscle Damage

Despite substantial research contributions to EIMD, the exact mechanisms responsible for damage, repair and adaptation from eccentric muscle actions remain inconclusive. A number of reviews have attempted to explain the mechanisms of EIMD and for a more detailed overview of these the reader is directed to previous literature.[1,9,19-25] This article further simplifies the damage model and has divided it into two general areas: (i) the initial phase or primary damage that occurs during the exercise bout; and (ii) secondary damage that propagates damage through processes associated with the inflammatory response.[26]

1.1 Primary Damage

The initial events that occur as a direct outcome of eccentric exercise may be further subdivided into two possible pathways: metabolic and mechanical.[1,20] Metabolic muscle damage has been proposed to result from ischaemia or hypoxia during exercise of a prolonged nature.[19,20] Ischaemia is thought to cause changes in ion concentration, metabolic waste accumulation and adenosine triphosphate deficiency, which ultimately result in damage[27,28] similar to that seen in EIMD.[29]

Some previous literature examining marathon running have reported the occurrence of damage and this may be a function of reduced metabolite clearance;[20] however, marathon running has a substantial eccentric contraction component and the associated mechanical stress may be the main cause of damage and any ischaemia may only exacerbate the damage from the eccentric contractions.[30] Comparisons of uphill and downhill running further support this, with uphill running showing a higher metabolic cost and no evidence of damage when compared with downhill running, which displayed a high incidence of damage at a lower metabolic cost.[31] More recently, Beltman et al.[7] showed a reduced metabolic cost during electrically stimulated lengthening contractions when compared with concentric and isometric contractions in rats. Therefore, metabolic factors seem to be unlikely candidates as a basis for EIMD occurring from eccentric biased exercise and may be limited to exercise of a long duration.

The mechanical hypothesis relates to the damage that occurs as a direct consequence of the mechanical loading on the myofibres. Eccentric contractions are capable of generating more force than isometric and concentric contractions and require less energy expenditure per unit of torque;[4,31-33] furthermore, lengthening of sarcomeres is non-uniform under eccentric conditions, which results in some myofilaments being stretched and no longer able to overlap within the sarcomere.[34,35] Consequently, when the filaments are stretched beyond the point of overlap, due to sarcomere inhomogeneity, passive structures assume more tension and undergo what is termed ‘popping’[36,37] and results in Z-band streaming.[23,38] The excessive tension that is placed upon these passive structures (such as desmin, synemin and titin) from repeated eccentric contractions may cause failure of the structure and is manifested as a reduction in the muscle’s ability to generate force, which has been reviewed in detail elsewhere.[1,9]

1.2 Secondary Damage

The processes that follow the primary phase of damage appear to be initiated by a disruption of the intracellular Ca^{2+} homeostasis. An increase in the intracellular $[Ca^{2+}]$ is thought to derive from extracellular sources,[1] and consequently lead to further myofibrillar damage in skeletal muscle.[39,40] Eccentric exercise has been shown to lead to a loss of sarcoplasmic reticulum membrane integrity and flux of Ca^{2+} into intracellular areas in rats, which culminates in skeletal muscle damage.[41,42] In addition, a recent contribution to the literature has shown that there is a disruption in calcium homeostasis resulting from changes in the sarcoplasmic reticulum following lengthening contractions in humans.[43] The influx of the Ca^{2+} into the cytosol initiates a cascade of events that further damage the cell by causing alterations to the cytoskeleton, sarcoplasmic reticulum, mitochondria and myofilaments.[40,44] The Ca^{2+}-mediated responses appear to be manifested by the activation of proteolytic and lipolytic pathways that lead to degradation of the cell membrane and sarcolemma, cell infiltration and subsequent activation, production of reactive oxygen species, fibre necrosis[9,45] and ultimately regeneration of the fibres some days later.[46] The loss of membrane integrity allows for the 'leakage' of intramuscular proteins, both structural and cytosolic, which can be seen in blood for many days post-exercise.[10] However, there has recently been evidence to show that necrosis may not occur after damaging eccentric exercise in humans.[47] This work and subsequent investigations from this group suggest that the tissue undergoes adaptation and regeneration rather than necrosis and therefore areas of increased cellular activity reflect synthesis and remodelling.[17,18,48]

The exact mechanisms that underlie the initial phases in muscle damage are not completely understood and it may be possible that initial damage is a function of anomalies in excitation-contraction uncoupling[49,50] or that passive structures within the sarcomere are compromised,[9,35,51,52] but there are most likely contributions from both areas.[9,49,51] There are convincing data to suggest that the exercise-damage-repair process is associated with the primary phase and is further exacerbated by the secondary phase,[53,54] although the extent to which each is responsible remains debatable.

1.3 The Occurrence of Damage

Investigations examining direct evidence of damage immediately post-exercise have provided convincing data of myofibrillar disruption resulting from eccentric contractions,[31,55,56] which appear to be focused within fast-twitch fibres, although more recent evidence suggests that some damage also occurs to some slow-twitch fibres.[57] Furthermore, there is also evidence that damage may be exacerbated during faster eccentric contractions, which may be due to reduced crossbridge formation that occurs at higher speeds and thereby increasing the force per crossbridge and causing failure of the contractile apparatus.[58] However, eccentric contractions do not have to be maximal in nature to elicit damage,[59-61] although more damage tends to occur when exercise is maximal and is carried out at longer muscle lengths.[62-65] The muscle generates excessive strain during lengthening, due to sarcomere inhomogeneities,[9] and the tension from within the muscle is derived to a larger extent from passive structures and less from active tension in the 'descending limb' of the length-tension relationship, consequently extensive myofibrillar disruption can occur.[31,38,66,67] Based on the current evidence, eccentric contractions, regardless of the speed or intensity of the contraction, tend to cause damage when the exercise bout is novel.

Eccentric muscle actions have been shown to be beneficial to a number of populations; they are integral to many sports and exercise activities and are often exaggerated by some exercisers and athletes in order to promote adaptation. Any intervention used either as a prophylactic or as a therapy that could help to attenuate the negative effects of these muscle actions could prove beneficial for pain reduction, maintaining training intensity and exercise adherence. This review will consider human studies that have investigated interventions to attenuate the negative effects of EIMD.

2. Preventative and Therapeutic Interventions to Attenuate Exercise-Induced Muscle Damage

2.1 Antioxidants

The initial events caused by unaccustomed eccentric exercise that manifest muscle damage elicit an inflammatory response in which phagocytosis and neutrophil respiratory burst result in the production of reactive oxygen species (ROS), thereby imposing an oxidative stress upon the tissue.[68] These ROS have been implicated in the secondary damage that follows the primary mechanical disruption. However, the exact nature of the relationship between ROS production, EIMD and muscle soreness is unclear as highlighted in a recent review.[25] Dietary antioxidants have been proposed to reduce ROS production and therefore oxidative stress-imposed damage; the use of antioxidants to reduce symptoms associated with EIMD has been extensively reviewed.[69] Most of the research to date has focused on the effects of the antioxidant vitamins C (ascorbic acid) and E (tocopherol), which forms the basis of this article with studies summarized in table I.

Vitamin C has been reported to have antioxidant properties[84] and a number of studies have demonstrated attenuated muscle damage and soreness when high doses are administered prophylactically.[70,71] Supplementation with 3000 mg/day for 14 days prior to 70 eccentric contractions of the elbow flexors and for 4 days post-exercise has been shown to significantly reduce muscle soreness for the first 24 hours post-exercise; thereafter a non-significant reduction in soreness was evident until 96 hours post-exercise.[70] Furthermore, significant attenuation of the glutathione ratio, which is used as a marker of oxidative stress, was observed for 24 hours post-exercise, indicating that vitamin C was effective in reducing oxidative stress, which may have accounted for the reduced soreness. Whilst this study demonstrated a non-significant attenuation of creatine kinase (CK) between 48 and 96 hours post-exercise, there were no effects of supplementation on maximal isometric force or range of motion (ROM). This study supported the findings of Kaminsky and Boal,[71] demonstrating that 3000 mg/day administered for 3 days prior to and for 4 days post-exercise (15 minutes of cyclic plantar flexion and extension) decreased muscle soreness. In addition, Thompson et al.[72] have previously shown that supplementation with 400 mg/day for 12 days prior to 90 minutes of intermittent running significantly reduced muscle soreness and plasma interleukin (IL)-6, although there was no effect on CK, malondialdehyde (MDA) or muscle function.

In contrast, a recent study has shown supplementation with vitamin C 3000 mg/day for 3 days prior and 5 days following eccentric exercise to have no effect on strength loss, muscle soreness or ROM.[73] Furthermore, some studies have found supplementation with vitamin C to be contraindicated in the reduction of EIMD. Increases in CK and lactate dehydrogenase (LDH), indicative of reduced membrane integrity, have been reported with vitamin C 12.5 mg/kg and *N*-acetyl-cysteine (NAC, which increases the intracellular antioxidant glutathione) 10 mg/kg administered for 7 days following 30 eccentric contractions of the elbow flexors.[74] In addition, supplementation with vitamin C 1000 mg 2 hours before and for 14 days following downhill running delays the recovery of muscle function despite being effective in elevating plasma ascorbate and attenuating the increase in MDA and consequent ROS.[75]

Vitamin E is another antioxidant that has been suggested to reduce intramuscular protein efflux by increasing membrane integrity.[85] Reduced CK concentrations in blood following muscle damaging exercise when supplementing with vitamin E have been reported.[76,77] These studies used prophylactic doses of 1000 IU/day for 12 weeks and 1200 IU/day for 14 days, respectively, demonstrating attenuated CK responses following damaging exercise. In contrast, however, another study found that despite vitamin E 1200 IU/day taken for 30 days prior to 240 maximal eccentric contractions manifesting a 2.8-fold increase in serum vitamin E concentration, there was no effect on serum CK, Z-band disruption, muscle torque or muscle soreness.[78] It is possible, therefore, that longer-term supplementation periods (i.e. >30 day) may be required to confer a protective effect on muscle. The effects of vitamin E supplementation on muscle damage and soreness have been recently reviewed.[86]

A number of studies have investigated the synergistic effects of combined vitamin C and E supplementation. Shafat et al.[79] reported that a dosage of vitamin C 500 mg/day and vitamin E 1200 IU/day taken for 37 days attenuated the decline in eccentric torque during 300 maximal eccentric contractions of the knee extensors and in muscle function for 2 days post-exercise compared with a control. Furthermore, females taking a daily dose of vitamin C 1000 mg and vitamin E 400 IU for 14 days prior to and for 2 days following eccentric exercise of the elbow flexors has demonstrated decreased blood protein carbonyls and MDA, indicative of reduced oxidative stress and damage.[80] These findings contradict recent work[81] in which prior supplementation with vitamin C 1000 mg/day and vitamin E 300 mg/day

Table I. Studies reporting effects of antioxidant supplementation on markers of exercise-induced muscle damage

Study	Supplementation	Exercise protocol	Effects post-exercise (compared with control group)
Bryer and Goldfarb[70]	Vit C (3000 mg/d) for 14 d prior and 4 d post-exercise	70 elbow flexor eccentric contractions	↓ CK ↓ DOMS ↓ Glutathione ratio ↔ Muscle force ↔ ROM
Kaminsky and Boal[71]	Vit C (3000 mg/d) for 3 d prior and 4 d post-exercise	15 min cyclic plantar flexion and extension	↓ DOMS
Thompson et al.[72]	Vit C (400 mg/d) for 12 d prior	90 min intermittent shuttle running	↔ CK, Mb ↔ MDA ↔ Muscle force ↓ DOMS ↓ IL-6 ↔ CRP
Connolly et al.[73]	Vit C (3000 mg/d) for 3 d prior and 5 d post-exercise	40 elbow flexor eccentric contractions	↔ DOMS ↔ Muscle force ↔ ROM
Childs et al.[74]	Vit C (12.5 mg/kg/d) and NAC (10 mg/kg/d) for 7 d post-exercise	30 elbow flexor eccentric contractions	↑ CK ↑ LDH ↔ DOMS ↔ ROM ↔ IL-6
Close et al.[75]	Vit C (1000 mg/d) 2 h prior and 14 d post-exercise	30 min downhill running	↓ MDA ↓ Muscle force ↔ DOMS
Sacheck et al.[76]	Vit E (1000 IU/d) for 12 wk prior	45 min downhill running	↓ CK in younger men ↓ iPF(2α) in older men
McBride et al.[77]	Vit E (1200 IU/d) for 14 d prior	Whole-body resistance exercise	↓ CK ↔ MDA ↔ DOMS
Beaton et al.[78]	Vit E (1200 IU/d) for 30 d prior exercise	240 knee flexor and extensor eccentric contractions	↔ CK ↔ Muscle force ↔ Z-band disruption ↔ DOMS
Shafat et al.[79]	Vit C (500 mg/d) and vit E (1200 IU/d) for 30 d prior and 7 d post-exercise	300 knee extensor eccentric contractions	↓ Decline in torque during exercise ↑ Muscle force ↔ DOMS
Goldfarb et al.[80]	Vit C (1000 mg/d) and vit E (400 IU/d) for 14 d prior and 2 d post-exercise	48 elbow flexor eccentric contractions	↓ MDA ↓ Plasma protein carbonyls ↔ Glutathione status
Mastaloudis et al.[81]	Vit C (1000 mg/d) and vit E (300 mg/d) for 6 wk prior	50 km ultramarathon run	↔ CK ↔ LDH ↔ Muscle force
Petersen et al.[82]	Vit C (500 mg/d) and vit E (400 mg/d) for 14 d prior and 7 d post-exercise	90 min downhill running	↔ IL-6 ↔ CK ↔ Lymphocytes ($CD4^+$, $CD8^+$, NK)
Jakeman and Maxwell[83]	Vit C (400 mg/d) or vit E (400 mg/d) for 21 d prior and 7 d post-exercise	60 min box stepping	Vit C: ↓ LFF, ↔ muscle force Vit E: ↔ LFF, ↔ muscle force

CK = creatine kinase; **CRP** = C-reactive protein; **DOMS** = delayed-onset muscle soreness; **IL-6** = interleukin-6; **iPF(2α)** = iso-prostaglandin 2α; **LDH** = lactate dehydrogenase; **LFF** = low frequency fatigue; **Mb** = myoglobin; **MDA** = malondialdehyde; **NAC** = *N*-acetyl-cysteine; **NK** = natural killer cells; **ROM** = range of motion; **vit C** = vitamin C (ascorbic acid); **vit E** = vitamin E (tocopherol); ↓ indicates decrease; ↑ indicates increase; ↔ indicates no change.

for 6 weeks had no effect on markers of muscle damage (LDH, CK) or hamstring or quadriceps function following a 50-km ultra-marathon. It is possible that this difference in findings reflects disparities in the nature of prolonged running and eccentric specific contractions; however, this study did demonstrate significant increases in markers of muscle damage, thereby justifying comparison. The role of ROS in perpetuating the inflammatory response has recently been challenged.[82] Petersen et al.[82] showed that supplementing 500 mg/day of vitamin C and 400 mg/day of vitamin E for 14 days prior to, and for 7 days following 90 minutes of downhill running increased plasma vitamin concentrations; however, there was no response on proliferation of cytokine or lymphocyte subpopulations.

Only one study has directly compared the effects of vitamins C and E in attenuating muscle damage.[83] This work compared supplementation of vitamin C 400 mg/day and vitamin E 400 mg/day for 21 days prior to and 7 days following 60 minutes of box-stepping exercise. Vitamin C significantly attenuated the decline in the 20/50 Hz ratio following exercise (this represents reduced low frequency fatigue, which may be a consequence of attenuated muscle damage) and a trend for greater recovery in muscle force in the 24-hour period post-exercise; no effects were found for vitamin E. A recent contribution to the literature has examined the impact of tart cherry juice on the attenuation of symptoms of exercise-induced muscle damage.[87] A daily dose of 0.682 L (24 fl oz) for 4 days prior and 4 days following eccentric contractions of the elbow flexors reduced muscle soreness and attenuated the loss in muscle force. Although this is the only study to have examined this supplement, the findings are promising and suggest a requirement for further research, not least to elucidate the mechanisms by which any benefit may be conferred.

In summary, the literature is equivocal with respect to the effects of antioxidant supplementation on EIMD. Differences in dose, supplementation period and exercise interventions adopted in previous studies confound the drawing of definitive conclusions; however, it is suggested that long-term supplementation (≥14 days) with either vitamin C or E alone, or in combination, may reduce signs and symptoms of EIMD through the reduction of ROS following muscle damage.

2.2 Carbohydrate and Protein

Muscle glycogen resynthesis has been shown to be impaired following high-intensity eccentric exercise.[87] Compared with a contralateral limb control, significantly lower glycogen resynthesis in the quadriceps was identified following eccentric knee extensor contractions followed by 60 minutes of cycling at 70% maximal oxygen uptake ($\dot{V}O_{2max}$).[88] It was concluded that the lower carbohydrate uptake in the eccentrically exercised limb was possibly due to inflammatory cells competing with muscle cells for blood glucose; however, high dietary carbohydrate for 3 days after eccentric exercise did increase intramuscular carbohydrate storage.[88] These observations have been supported by subsequent studies.[89,90] When eccentric contractions were performed following glycogen-depleting exercise (comprising 6 × 15-second sprints and 1-hour submaximal ruining) not only was muscle glycogen reduced further in the 2 hours following exercise, muscle glycogen resynthesis was also significantly attenuated over the subsequent 48-hour period.[88] This supports a previous investigation,[90] which also identified that the deficit in glycogen content peaked at 72 hours post-exercise, which may coincide with the delayed inflammatory response that occurs after the initial myofibrillar disruption from damaging exercise. Close et al.[91] compared high-carbohydrate (77%) and low-carbohydrate (11%) diets adopted for 48 hours prior to 30 minutes of downhill running. There was no significant difference in muscle soreness, CK, MDA, glutathione or muscle function; interestingly, however, the high-carbohydrate diet demonstrated trends for higher soreness and CK at 24 and 48 hours post-exercise. Further evidence refuting the efficacy of carbohydrate in attenuating muscle damage has been presented[92] with no differences between glycogen-depleted and glycogen-repleted states for muscle soreness, maximal isometric force, relaxed knee angle and thigh circumference following 15 minutes of downhill running.

There is very little research that has examined the role of protein supplementation in preventing or alleviating symptoms of EIMD. Shimomura et al.[93] recently reported that a 5-g supplement of branched-chain amino acids administered 15 minutes prior to squat exercise reduced muscle soreness for 4 days post-exercise in women, but not men. It is possible that the discrepancy between sexes was attributable to the lower relative dose in the men, as suggested by the authors. Another recent study[94] compared the effects of two amino acid supplementation strategies. Whilst supplementation with 3.6 g taken 30 minutes prior and immediately following exercise (900 arm-curl repetitions) had no effect on any markers of muscle damage, when supplementation was extended for 4 days post-exercise (providing a total dose of 36 g) there was a significant reduction in CK, blood myoglobin and muscle soreness. It is clear that further research is required to elucidate the role of protein supplementation in the prevention and treatment of EIMD.

The co-ingestion of protein and carbohydrate may also provide synergistic effects in attenuating muscle damage.[95] Using a protocol of repeated-cycle trials to exhaustion (the first at 75% peak oxygen uptake [$\dot{V}O_{2peak}$], the second at 85% $\dot{V}O_{2peak}$ 12–15 hours later), compared with carbohydrate alone, the ingestion of a carbohydrate-protein beverage during and immediately after exercise significantly attenuated CK concentrations and significantly improved performance (106.3 ± 45.2 and 43.6 ± 12.5 minutes vs 82.3 ± 32.6 and 31.2 ± 8.7 minutes, for carbohydrate-protein and carbohydrate trials, respectively). These findings indicate that the co-ingestion of carbohydrate and protein increases time to exhaustion and reduces muscle damage in prolonged endurance exercise. Small reductions in CK levels following eccentric exercise (100 quadriceps eccentric contractions) performed in a glycogen-depleted state have been reported as a result of ingesting a carbohydrate-protein mix 2 hours and immediately prior to exercise;[96] however, in contrast, this study found no effect on other markers of muscle damage (muscle function, plasma IL-6, or urinary-3 methyl-histidine).

Whilst there is evidence that muscle glycogen resynthesis is impaired by exercise-induced muscle damage, it appears that carbohydrate administration has little or no effect in attenuating signs and symptoms of muscle damage. There is, however, some evidence that protein supplementation and the simultaneous ingestion of carbohydrate and protein may confer protection against exercise-induced muscle damage; whether the co-ingestion of protein and carbohydrate provides any benefit over the ingestion of protein alone is a question that has not yet been investigated and warrants research.

2.3 β-Hydroxy-β-Methylbutyrate

β-Hydroxy-β-methylbutyrate (HMB) has recently gained popularity as a dietary supplement in humans, particularly among strength athletes.[97] HMB is a metabolite of the branched-chain amino acid leucine and its ketoacid, α-ketoisocaproate,[98] and has been proposed to have a prophylactic effect in attenuating muscle protein degradation and exercise-induced damage during intense physical training. Although no direct evidence exists as to exact mechanisms for these effects, HMB has been hypothesised to act as a precursor for cholesterol synthesis via its metabolism to β-hydroxy-β-methylglutaryl CoA (HMG-CoA), thus providing a carbon source for cholesterol synthesis.[99] An increase in intracellular cholesterol could enhance cell membrane integrity and therefore reduce muscle damage following unaccustomed or intense exercise.[99] Another hypothesis is that HMB serves as a structural component within the cell membrane.[100]

Supplementation with dosages of between 1.5 and 3.0 g/day has been shown to reduce muscle protein degradation with consequent increases in muscle mass and strength following resistance exercise training performed for between 3 and 8 weeks;[100-103] however, there is some evidence that these benefits are not observed in well trained subjects.[104,105] In addition, blood CK activity has also been shown to be reduced by HMB supplementation following resistance training[100-103] and prolonged running.[106]

To date, only two studies have examined the effects of HMB supplementation on muscle damage resulting from a single bout of eccentric resistance exercise. Paddon-Jones et al.[107] found that 6 days of HMB supplementation with 40 mg/kg body mass/day provided no attenuation of symptoms associated with exercise-induced muscle damage following 24 maximal eccentric contractions of the elbow flexors. In contrast, longer-term supplementation (3 g/day for 14 days) prior to a single bout of eccentric resistance exercise has been reported to attenuate the decrement in muscle function, the CK response and limb swelling.[108] The evidence therefore suggests that supplementation with HMB for >6 days prior to exercise may attenuate symptoms of EIMD; however, it should be noted that this conclusion is based on studies examining responses in relatively untrained individuals who are unaccustomed to eccentric exercise.

2.4 NSAIDs

NSAIDs, e.g. ibuprofen, aspirin (acetylsalicylic acid), naproxen, diclofenac, flurbiprofen and ketoprofen, are perhaps the most widely known therapy in the treatment of muscle damage and delayed-onset muscle soreness (DOMS). NSAIDs inhibit the synthesis of prostaglandin, a potential mediator of oedema and pain during acute inflammation, by inhibiting the metabolism of arachidonic acid via the cyclo-oxygenase (COX) pathway.[109-114] Many studies have investigated the effects of NSAIDs in animal models; however, this review focuses only on studies in humans. The prophylactic and therapeutic use of NSAIDs to alleviate the symptoms of EIMD has been recently reviewed.[115]

Prophylactic administration of ketoprofen has been shown to significantly attenuate soreness and enhance the recovery of muscle function following maximal eccentric exercise;[112] with doses of 25 and 100 mg reducing soreness by 19% and 10%, respectively, and enhancing muscle function by 9% and 16%, respectively. Similarly, 150 mg diclofenac sodium ('Voltaren') administered orally for 27 days reduced muscle inflammation and CK following 20 minutes of strenuous stepping performed on the 15th day of supplementation, thus indicating reduced muscle damage.[116] Furthermore, ibuprofen taken before and after exercise (2400 mg) attenuates muscle damage as demonstrated by reduced CK appearance in blood.[117] Performance benefits of NSAID administration are, however, less clear. Tokmakidis et al.[118] prescribed 400 mg ibuprofen every 8 hours for 48 hours following eccentric leg-curl exercise; although there was significantly less muscle soreness at 24 hours and CK at 48 hours, there was no effect on maximum strength, vertical jump performance or knee ROM. In contrast, some studies have found NSAIDs to have no effect on inflammation or muscle soreness. Donnelly et al.[119] reported that doses of 1200 mg ibuprofen administered prior to 45 minutes of downhill running, together with a further 600 mg every 6 hours for 72 hours post-exercise, had no effect on muscle soreness or strength; however, higher CK responses were measured following supplementation. Gulick et al.[120] found 1200 mg of oxaprozin taken every 24 hours for 72 hours post-exercise had no effect on indices of muscle damage. Similarly, Peterson et al.[121] used over-the-counter maximum doses of ibuprofen (1200 mg/day) and paracetamol (acetaminophen) [4000 mg/day] following damaging exercise with no effect on the inflammatory response 24 hours post-exercise.

Evidence is currently equivocal for the use of NSAIDs in the prevention and/or reduction of the signs and symptoms associated with EIMD. The use of higher doses of NSAIDs in future research has been advocated,[112,122] although caution should be exercised as the habitual or long-term use of NSAIDs may produce adverse effects causing further medical complications.[114,115,123] Furthermore, inflammation is the body's natural response to EIMD and the use of pharmacological agents may inhibit protein synthesis;[124] thus inhibiting the damage-repair-adaptation cycle of skeletal muscle to eccentric exercise. Evidence of suppressed tissue protein synthesis following high-intensity eccentric exercise as a result of over-the-counter doses of ibuprofen and paracetamol administration (1200 mg/day and 4000 mg/day, respectively) has recently been shown.[125] Indeed, there is evidence of inhibited muscle regeneration and hypertrophy following NSAID administration in animal models.[126,127] Given the equivocal acute effects of NSAIDs, the lack of performance benefits reported, and the negative long-term consequences of administration, the use of NSAIDs is not recommended as an effective strategy to prevent or treat symptoms of EIMD.

2.5 Stretching

Stretching is proposed to decrease the passive or active stiffness of skeletal muscle, making it more compliant to eccentric contractions and thereby reducing the amount of primary mechanical damage. Accordingly, pre-exercise static stretching has been suggested as a possible intervention to attenuate the magnitude of muscle damage.[128] A brief meta-analysis of five studies[129] found stretching to reduce soreness in the 72 hours following exercise by 2 mm on a visual analogue scale of 100 mm, but suggested that this magnitude of change is too small to be worthwhile. LaRoche and Connolly[130] reported that long-term stretching (4 weeks of static or ballistic stretching) maintained ROM and stretch tolerance in the days following eccentric exercise, but had no effect on muscle soreness. Pizza et al.[131] have shown that passive stretching performed prior to eccentric exercise reduces the inflammatory response, thus suggesting that prior stretching may protect the muscle cell against exercise-induced muscle damage. Indeed, Koh and Brooks[132] have reported passive stretching to provide protection against muscle damage when damaging exercise is performed 2 weeks later; furthermore, the magnitude of this protection was approximately 50% of that afforded by prior eccentric exercise (i.e. the repeated-bout effect [RBE], as discussed in section 2.10). Reduced CK activity and attenuated losses in force have been shown when stretching is performed in combination with warm-up and post-exercise massage.[133]

In contrast, static stretching performed prior to eccentric exercise has been shown to no have no effect on muscle soreness and maximal force[134,135] or the CK response.[135] It is important to note that both of these studies used female participants; it is therefore possible that sex difference may account for these findings. It has been widely reported that estrogen provides protection against damage in skeletal muscle and that females respond to damaging exercise in a different manner to males, dependent upon menstrual cycle phase.[24] However, High et al.[136] have also reported static stretching to have no effect on muscle soreness in a mixed sex sample following eccentric exercise; this study compared stretching, warm-up and stretching with warm-up protocols, with no intervention having any effect. In contrast, by employing five rapid arm extensions following eccentric exercise, passive stiffness and muscle soreness has been shown to be temporarily reduced; however, passive stiffness was seen to increase over the following hour.[137]

Current literature reporting the efficacy of stretching to prevent or reduce EIMD suggests only minimal effect in reducing muscle soreness and no effect on performance. It has, however, been previously highlighted that other as yet un-investigated stretching protocols, such as proprioceptive neuromuscular facilitation, ballistic or dynamic stretching techniques may be of benefit, and therefore present a direction for future research.[129]

2.6 Massage

Massage is a widely used therapy in the treatment of athletic muscle soreness and micro-injury and many athletes are convinced by its potential to alleviate muscle soreness. Tiidus[138] highlighted a lack of consensus in the literature as to the effects of and mechanisms by which massage might be effective; however, there is now some literature base to support its use in the treatment of exercise-induced muscle soreness.

Massage performed 2 hours following eccentric exercise has shown significant reductions in muscle soreness and CK appearance in blood.[139] These effects may be explained by the physical disruption of neutrophil accumulation within muscle and a consequent reduction in prostaglandin production and pain.[139] Similar findings have been reported for 10 minutes of deeply applied clearing techniques 3 hours post-exercise, significantly reducing muscle soreness, CK and limb circumference; however, no effects were found for muscle strength or ROM.[140] These findings confirm those of Hilbert et al.,[141] who also found reduced soreness, but no change in muscle function (strength or ROM) when 20 minutes of massage was applied 2 hours following maximal eccentric exercise; however, in contrast to the conclusions of Smith et al.,[139] no reduction in neutrophil count was observed.[141] Performance benefits have, however, been demonstrated alongside reduced muscle soreness when female collegiate volleyball and basketball players were treated with effleurage, petrissage and vibration massage during pre-season training, 2 days after intense training commenced.[142] As previously highlighted, massage provided in combination with warm-up and stretching can also attenuate losses in force following eccentric exercise.[133]

The majority of the evidence to date points towards massage being effective in alleviating muscle soreness, although its effect on muscle function and performance is less clear. This may in part be due to a number of methodological limitations in the literature, (e.g. inadequate therapist training, different massage techniques used, insufficient duration of treatment) that have been highlighted in previous reviews.[143,144]

2.7 Electrotherapeutic Modalities

Electrical modalities have been used in the treatment of injury and are extensively utilized for the rehabilitation of a range of musculoskeletal problems; however, the use of electrical therapies to reduce symptoms associated with muscle damage has not been extensively examined.

Trancutaneous electrical nerve stimulation (TENS) is used for pain control and to elicit an anti-inflammatory response.[122] Denegar et al.[145] examined pulsed TENS application in females experiencing DOMS, demonstrating a significant attenuation of pain and the reduction in ROM. These effects were attributed to the ‘Pain Gate Control Theory’, which suggests that presynaptic pain fibres are inhibited by stimulation of large sensory fibres in the muscle.[146] Such findings are supported by those of Denegar and Perrin,[147] demonstrating that TENS applied 48 hours following eccentric exercise significantly reduced muscle soreness in females. In addition, this study identified that whilst TENS had no effect on ROM, when applied in conjunction with cold application ROM was increased; TENS had only a non-significant effect on muscle strength.

A similar modality to TENS, microcurrent electrical neuromuscular stimulation, however, appears to have no effect on muscle function or pain when applied immediately after and at 24 hours post-exercise[148] and no effect on muscle soreness or ROM when applied at 24, 48 and 72 hours post-exercise.[149] In contrast, electro-membrane microcurrent therapy significantly attenuates the increase in CK and the decrease in ROM and muscle strength at 24 hours; although perhaps surprisingly, has no effect on muscle soreness.[150] The mechanisms for these observations are not known, although it is proposed that disturbances in calcium homeostasis that are inherent with eccentric exercise may be reduced by such treatment.[150]

Another electrotherapeutic modality examined is high-voltage pulsed current (HVPC), which is applied via a short pulse duration monophasic waveform directly through electrodes. HVPC has been demonstrated in animal models to alleviate muscle soreness by reducing oedema accumulation; however, the only two studies that have examined HVPC in humans[151,152] have shown no effects on muscle soreness or function (strength and ROM).

Ultrasound converts electrical energy into sound waves that penetrate the skin and is proposed to increase circulation, accelerate metabolism and control the sensation of pain.[153] There is evidence that ultrasound treatment applied for 20 minutes at 24 hours following stepping exercise significantly reduces muscle soreness.[154] Ultrasound therapy and the administration of an anti-inflammatory cream may have synergistic effects, with significantly less muscle soreness reported when both treatments were used when compared with ultrasound alone.[155] It was proposed that the combined treatment allowed for increased blood flow to the damaged tissue whilst reducing the inflammatory response and consequent oedema and secondary damage. The paucity of experimental research investigating the effects and mechanisms of ultrasound therapy has been previously highlighted.[138]

In summary, electrical therapies show some promise in the treatment of EIMD and muscle soreness; however, these electrical mediums require specialist training to administer and the equipment itself can be expensive. There are currently no evidence-based guidelines on the dose and frequency of application concerning electrical therapies. Robertson[156] has stated that guesswork is currently used in the dose-response treatment using ultrasound. Therefore, further research is required to elucidate the efficacy of electrical therapies using standardized guidelines based on documented evidence. However, given that these therapies target isolated muscles, the application of such findings may be of limited value in an athletic environment, in which individuals rarely experience such isolated muscle damage and soreness.

2.8 Cryotherapy

Cryotherapy is the application of cold for therapeutic purposes. It is usually applied during the acute stage of trauma and is proposed to diminish the undesirable effects of soft tissue injury[111] by reducing the inflammatory response, swelling, oedema, haematoma and pain.[157-160] The therapeutic effects are proposed to be a consequence of analgesia of the injured tissue, hypometabolism of the tissue, and a vascular response whereby blood flow and haematoma formation is retarded.

Single treatments of ice massage appear to have very limited efficacy in treating muscle damage and soreness, with only a short-term reduction in DOMS when ice massage was applied immediately following exercise; importantly, however, this was mirrored by a reduction in muscle force.[120] Similarly, single ice massage applications of 15 minutes applied either immediately post-exercise, or at 24 or 48 hours post-exercise have no effect on DOMS or ROM in females following eccentric exercise.[161] These studies indicate that single treatments of ice massage provide limited, if any, alleviation of DOMS.

Repeated applications of ice massage following eccentric exercise have demonstrated non-significant decreases in muscle strength and ROM and non-significant increases in muscle soreness and CK, thereby identifying possible contraindications of ice massage in the treatment of EIMD.[162] This study was, however, limited by sample sizes of only 5–6 participants in treatment groups. Ice massage administered immediately after and at 24-hour intervals for 48 hours following eccentric exercise has been shown to attenuate CK efflux at 72 hours post-exercise,[163] but to have no effect on muscle soreness or function.[163,164]

Repeated cryotherapy application in the form of cold water immersion has also been unsuccessful in alleviating muscle soreness and EIMD. Paddon-Jones and Quigley[165] adopted a protocol of 20-minute cold water immersions (5 ± 1°C), separated by 60 minutes and repeated five times following eccentric exercise; it is possible that such an aggressive approach may have elicited the Hunting Reaction. This phenomenon is brought about when tissue temperature falls below 18°C and there is periodic vasodilation and warming of the tissue, which may eliminate any potential benefits of cryotherapy. A less intense treatment regimen involving immersion in water of 15°C for 15 minutes, every 12 hours for 3 days following eccentric exercise resulted in a significantly greater relaxed elbow angle (indicative of reduced passive stiffness) and lower CK levels in blood, but with no effects on muscle soreness or strength.[166] Beneficial effects of cold water immersion following eccentric exercise have also been reported by Yanagisawa et al.[167] A single treatment immediately post-exercise and repeated treatments immediately following and at 24 hours post-exercise of 15 minutes at 5°C both significantly reduced soreness at 48 hours post-exercise and the increase in magnetic resonance-T2 weighted image, which is a measure of relaxation time, at 96 hours post-exercise. A recent study has examined the practice of intermittent ice-water immersion, which has become a popular practice amongst athletes in many countries.[168] Immersion in ice-water (5°C) repeated three times and interspersed with 1-minute periods out of the water resulted in no changes in CK, muscle function or swelling; however, the intervention did result in significantly greater muscle pain.

Cryotherapy appears an attractive strategy to combat the signs and symptoms of EIMD because it offers a quick, simple and if required, whole-body, intervention. Indeed, there are many anecdotal reports of the widespread use of part or whole-body cold water immersion following competition and training by athletes in an attempt to enhance recovery. In contrast, however, the evidence regarding the effects of cryotherapy on muscle function and soreness is equivocal, with perhaps the more effective treatment strategies comprising conservative and repeated applications of cold media.

2.9 Exercise

Armstrong[19] has suggested exercise to be the most effective strategy to alleviate DOMS; however, it seems that the analgesic effect is only temporary[169] and pain tends to resume after exercise has stopped.[114] Indeed physical therapists prescribe light exercise in the presence of muscle soreness; however, the appropriate mode, intensity and duration have yet to be elucidated.[170] Hough[171] purported that increased blood flow removed noxious waste products and increased endorphin release, thereby causing an analgesic effect. However, the research-based evidence pertaining to the efficacy of exercise as a treatment in attenuating symptoms of EIMD remains equivocal.[84,114]

Nosaka and Clarkson[172] showed that fatiguing concentric exercise conducted immediately prior to eccentric muscle actions significantly attenuated damage markers. One might expect that the fatiguing concentric exercise would affect the production of eccentric force; however, the eccentric torque during the damaging exercise was similar to the control group. A more recent investigation[173] examining the prophylactic effects of warm-up on markers of muscle damage compared groups of low-heat passive warm-up, high-heat passive warm-up, active warm-up, and no warm-up prior to eccentric exercise and a control. The investigators concluded that passive warm-up may be more beneficial than an active warm-up in reducing swelling; however, it does not attenuate any other damage indices or resolve symptoms of EIMD.[173] These investigations have a methodological design that make the application to athletic and exercising populations difficult; consequently they appear to lack external validity. Perhaps a protocol with more external application should be employed in future investigations to elucidate the prophylactic effects of prior exercise.

Hasson et al.,[174] who examined therapeutic exercise, showed that high-speed voluntary contractions carried out 24 hours after eccentric stepping exercise had some positive effects. Despite demonstrating less soreness and an attenuation of torque at 48 hours post-exercise, it is not clear why these high-speed contractions may be of benefit when other investigators using exercise interventions[172,173,175] have not.

A recent addition to the literature[175] had subjects complete one of three conditions (passive recovery, running at 50% $\dot{V}O_{2max}$, and electromyostimulation carried out at 30 minutes, 24, 48 and 96 hours post-exercise). Each trial was separated by 3 weeks, in a randomized, crossover design. No differences were found between conditions; however, these data should be treated with some caution as this experimental design is likely to be confounded by the fact that all subjects completed all conditions and there was insufficient provision to account for the RBE, which has been shown to last for >12 weeks[176,177] and is discussed in section 2.10. It would appear that exercise conducted as a prophylactic intervention or as a therapy is of limited value in attenuating symptoms associated with EIMD, and therefore other areas are perhaps more worthwhile to explore.[178] Having said this, there are some limitations within the literature that could be addressed in order to elucidate the value of exercise administered before and/or after potentially damaging exercise.

2.10 Repeated Bout Effect

The signs and symptoms of EIMD following eccentric exercise can be greatly attenuated following a second bout of a similar magnitude. This protection or adaptation is commonly referred to as the RBE[26,179,180] and has been demonstrated in both animals[30,181-183] and humans using the lower limbs[56,184-190] and the upper limbs.[179,191-194] The RBE not only provides protection that seems to occur within a few days,[179,194-196] but has also been reported to last for several months in humans.[176,177]

The damage response has a tendency to be greater in the upper limb than the lower limb, probably due to the lower limb being exposed to regular prior bouts of eccentric exercise for day-to-day locomotion such as descending stairs. The upper limb is non-weight bearing and most individuals are unaccustomed to high-intensity eccentric loading using this limb. Upper-limb exercise may lack specificity when making inference to many sports and exercise, but can provide a valuable insight into the damage-repair-adaptation response of human skeletal muscle to EIMD. Several theories have been mooted as potential mechanisms for the RBE; these include mechanical, cellular, neural, enhanced excitation-contraction coupling as well as a growing body of evidence of an altered inflammatory response and/or change in reactive oxidative species during and after the repeated bout; for a concise review of these possible theories, the reader is directed to McHugh.[26] Our review uses examples of the RBE and focuses on more recent contributions to the literature that provide information on factors that influence the protection of skeletal muscle from a single bout of eccentric contractions.

Early investigations have clearly demonstrated the RBE by showing an attenuation of indirect damage indices such as muscle soreness, CK and muscle function following the repeated bout using upper-limb[197] and lower-limb models.[184,195] Direct evidence for the RBE has also been demonstrated;[56] biopsy revealed reduced focal damage to the myofibrils in two-thirds of subjects at day two following the repeated bout and all subjects showed minimal soreness, CK elevations and decrement in muscle function. More recent investigations support earlier work concerning the RBE and demonstrate compelling and conclusive evidence of an attenuation of the signs and symptoms of EIMD following a repeated bout or eccentric exercise.[177,189,197]

Most investigations allow sufficient time for markers of damage to return to pre-exercise levels before exposing the muscle to a further insult of exercise, which usually occurs within 2 weeks. However, some investigators have examined the RBE prior to the recovery of soreness and function, which often occurs in sporting populations that may conduct several potentially damaging exercise bouts over a short epoch. Nosaka and co-workers have made considerable contributions to the literature in the field of muscle damage and adaptation. One such investigation examined whether repeated bouts of eccentric exercise would exacerbate damage when repeated 2 and 4 days after the initial bout. The repeated bouts did not exacerbate damage or impede the time to recovery, which has subsequently been supported by more recent work.[194] This is possibly due to focal damage occurring to stress-susceptible fibres during the initial bout and therefore further damage could not occur during the repeated bouts. In addition, the ability to generate force during bouts two and three was inhibited considerably; almost certainly as a result of the damage inflicted from the initial bout and therefore the force generated during subsequent bouts was insufficient to elicit any further damage.

Chen[194] examined repeated bouts of eccentric exercise 3 days after the initial bout. Subjects completed 30 maximal eccentric contractions followed 3 days later by 30 or 70 maximal eccentric contractions using the elbow flexors. Damage was evident in all groups and was not exacerbated after the second bout, even in the groups that completed more contractions in the repeated bout, which concur with the findings of Nosaka and Newton.[196] Furthermore, in Chen's investigation[194] the second bout was conducted when initial damage was still evident and less torque could be generated (38% less in the repeated bout of the 30 repetition group); therefore, it is perhaps unsurprising that no further damage occurred. From an applied standpoint, it seems that if the skeletal muscle has residual damage from eccentric contractions, then further insults of exercise are unlikely to incur further detriments in performance or inhibit the recovery process. However, if the purpose of further exercise bouts (prior to full recovery) is to induce greater adaptation, then the exercise stimulus from the repeated bout might be insufficient to induce additional benefits, although further research should be conducted to examine this possibility.

It is also possible that an adaptation or prophylactic effect on subsequent bouts may be manifested by an initial bout that causes only minimal damage and soreness. Protection is conferred from bouts of a low volume on subsequent bouts of higher volume eccentric contractions,[188,191,198,199] which suggests that the RBE may occur from a relatively low stimulus.

Clarkson and Tremblay[191] have shown that 24 maximal eccentric contractions of the forearm flexors (that induced modest damage) could attenuate damage from a subsequent bout of 70 maximal contractions. Since this investigation, others[188] have reported similar findings using the lower-limb model, where subjects performed 10, 30 or 50 maximal eccentric contractions followed by a second bout of 50 contractions, carried out 3 weeks later. The ten-repetition regimen caused appreciably less initial damage compared with 30 or 50 repetitions. However, all groups showed significant attenuation after the repeated bout, which demonstrates that a maximal low volume bout provides protection from maximal higher volume bouts. Likewise, others have further corroborated these findings and have showed evidence of a RBE after an initial bout of only 2 and 6 maximal eccentric contractions, using elbow flexors, prior to a repeated bout of 24 contractions[199] and most recently that ten maximal eccentric contractions provides protection from a bout of 45 maximal eccentric contractions conducted 2 weeks later;[198] although the magnitude of the protection appears to be less from a lower volume bout compared with a high volume bout.

Another variable that can be manipulated is exercise intensity. Nosaka and Newton[61] showed less damage in an initial bout of submaximal exercise when compared with maximal exercise. Although it would appear that whilst the initial bout does not have to cause appreciable damage (evident from reduced volume), it does have to be of a maximal intensity to confer protection against subsequent maximal bouts. Nosaka and Newton[200] examined the effects of 8 weeks of submaximal (50% isometric one repetition maximum) concentric or eccentric training on damaging maximal eccentric exercise carried out at the end of the training period. Both eccentric and concentric training failed to provide a protective effect; therefore, in order to confer protection from EIMD the exercise should be muscle group and intensity specific. Having said this, lower intensity eccentric exercise does appear to protect against subsequent bouts of a similar low volume and intensity.[193]

Muscle length during the initial bout of contractions may also affect the RBE. Eccentric muscle actions carried out at shorter muscle lengths, i.e. the ascending limb of the length tension curve, provide little[38] or no protection[65,201] against a subsequent bout at longer muscle lengths. This is probably because there is little or no initial damage at short muscle lengths due to a substantial overlap within the sarcomeres and hence active tension can be generated by contractile elements within the myofilaments. Consequently, there is less mechanical stress and adaptive stimulus imposed on structures responsible for passive tension, which occurs at longer muscle lengths. Therefore, exercisers would be prudent to consider this and conduct muscle actions over the entire anatomical range to maximize the prophylactic potential of the initial bout. Lastly, the adaptation itself has been reported to last from between 6 to 9 months,[176] although anecdotally, many strength and conditioning coaches report that athletes tend to experience muscular soreness and reduced function after far shorter lay-off periods between bouts. Nosaka et al.[177] have, however, suggested that some dependent variables have a reduced attenuation after 8 weeks, which is perhaps more representative of applied observations.

The RBE is a phenomenon that has been consistently demonstrated by various investigators using upper- and lower-limb models. This adaptive response to a single bout of eccentric exercise is probably the only intervention that has consistently shown a positive prophylactic effect in attenuating EIMD; although a number of factors should be considered when prescribing potentially damaging exercise, particularly when the exerciser is expected to complete further bouts of eccentric biased exercise. If the exercise bout is novel, it would seem prudent to start with a low-volume, high-intensity bout in order to: (i) reduce the negative effects of eccentric exercise after the initial bout to allow minimum interference with further training bouts; and (ii) to confer a RBE against a subsequent high-volume bout.

3. Conclusion

Eccentric muscle actions form an integral part of human locomotion and have been demonstrated to benefit a wide number of populations. Eccentric contractions require a lower metabolic cost per unit of torque compared with concentric and isometric contractions, making them an appealing mode of exercise for a variety of clinical and special populations. Furthermore, athletic populations employ eccentric contractions in order to elicit the stretch reflex and promote hypertrophy in skeletal muscle. When eccentric contractions are novel or unaccustomed, a series of events result in temporary muscle damage and soreness, which may negatively affect performance in subsequent bouts of exercise. In an attempt to attenuate these negative effects, a variety of interventions have been used as a prophylactic and/or a therapy; these include nutritional, pharmacological, electrotherapies, manual therapies and exercise. Many studies use isolated muscle to model the effect of treatments; consequently, much of the existing literature lacks external validity when applied directly to a specific sport activity, such as post-match recovery, and hence there is a need for further investigation. Prior bouts of eccentric exercise have been consistently shown to elicit a prophylactic effect on subsequent bouts of potentially damaging exercise; conversely, many other treatments have shown some promise, but the research evidence remains equivocal, which may be due to inconsistencies in the dose, frequency and intensity of the intervention. Very often there is little or no evidence-based practice in applying these interventions and hence it is difficult to ascertain an appropriate strategy to combat the negative effects of EIMD. Further research is warranted to elucidate the most appropriate dose, frequency and intensity of these interventions in order to determine 'best evidence-based practice'. Furthermore, it is unclear how these interventions affect the adaptation process and it might transpire that treatments are being prescribed that actually attenuate adaptation; therefore, there is a pressing need to ascertain the full implication of these interventions on their ability to reduce the negative effects of EIMD and the impact of prophylactic and therapeutic interventions on adaptation.

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